A-A-50963
March 21, 1984

COMMERCIAL ITEM DESCRIPTION

SOLDERING IRON, SELF HEATING (WITHOUT ELECTRIC OR EXTERNAL HEAT)

The General Services Administration has authorized the use of this commercial item description in preference to Federal Specification GGG-S-600.

This description covers soldering irons which are heated without any type of external energy applied to the soldering tip. The heat is furnished by the firing of a single-shot heating cartridge unit loaded with a chemical thermal charge.

Salient characteristics.

Classification. This description covers self heating soldering irons with the following tip styles and sizes. The quantity, style, and size tips required for a specific procurement shall be as specified in the contract or order.

Style A - Pyramid Tip

Size 1 - 1/8 inch tip diameter (3 mm)

Size 2 - 1/4 inch tip diameter (6.5 mm)

Size 3 - 3/8 inch tip diameter (9.5 mm)

Size 4 - 5/8 inch tip diameter (16 mm)

Style B - Chisel Tip

Size 1 - 1/4 inch tip diameter (6.5 mm)

Size 2 - 3/8 inch tip diameter (9.5 mm)

Size 3 - 5/8 inch tip diameter (16 mm)

Size 4 - 1 inch tip diameter (25.5 mm)

Design. The self-heating soldering iron covered by this specification shall be of the latest design, convenient to operate and entirely adequate for soldering wire connections, joining seams for large and small surface areas, and similar applications.

FSC 3439

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Body. The soldering iron shall have a body assembly manufactured of corrosion-resisting steel and shall be of sufficient strength and rigidity to support the forces exerted in normal soldering operations. The body shall support and guide the firing mechanism. Provisions shall be made for attaching the different styles of tips and for securing the handle to the body.

Handle. The soldering iron shall have a handle of thermosetting plastic which shall be securely fastened to the body assembly. The exterior of the handle shall be smooth and comfortable to the hand of the operator.

Reflector. The soldering iron shall have a reflecting disc approximately the diameter of the handle and manufactured from corrosion-resisting steel. The reflector shall be attached to the soldering tip end of the handle.

Firing pin mechanism. The soldering iron shall have a firing pin mechanism. The mechanism shall be self-contained and guided by the body assembly. The firing pin mechanism shall be designed and constructed to be manually discharged for release of the thermal charge to heat the iron.

Soldering tips. The soldering tips shall be manufactured from copper.

Heating units. The heating unit shall be a single shot, self-contained cartridge with a chemical thermal charge and percussion cap for priming or any other equally effective self-contained heating unit. It shall be designed for insertion in the body assembly in a manner to make proper contact with the firing pin. It shall supply the energy required for heating the soldering tip to a minimum temperature of 180 degrees C in 30 seconds or less and be capable of maintaining a soldering termperature for 1/8 inch diameter ASTM 50 general purpose solder with each style and size of tip being procurred, for a minimum of six minutes at an ambient temperature of 24 degrees C. At minus 40 degrees C the unit shall reach a temperature of 180 degrees C in 30 seconds or less and maintain a soldering temperature for a minimum of three minutes when tested as specified above. If the heating unit is of the chemical percussion priming type the cartridge shall be nonexplosive without a percussion contact primer and have an ignition point of 315 degrees C. The primer shall be inset from the plane of the cartridge head as a precaution against accidental discharge. Cartridges shall be packed 12 to a box. One box of cartridges shall be furnished with each soldering iron.

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Wire brush. One wire brush shall be furnished with each soldering iron. The brush shall be of appropriate diameter and material to remove residue from the soldering tip and cartridge of the body assembly.

Identification marking. Each iron shall be marked in a permanent and legible manner with the manufacturer's name or with a trade-mark of such known character that the source of manufacture may be readily determined.

Responsibility for inspection. Unless otherwise specified in the contract,the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the CID where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Acceptance inspection. The supplier shall randomly select the number of samples (soldering irons, cartridges, and tips) specified in Table I that is applicable to the quantity of items being purchased. Each sample shall be visually inspected, measured and tested by the supplier to determine conformance with the salient characterisitics specified herein. Any sample failing to meet these characteristics shall be considered a defective item. If the number of defective items equals, or is less than the quantity specified under "Acceptance, Number Defective" in column of Table I, the supplier shall replace the defective samples with acceptable items and the entire lot shall be considered acceptable for shipment. If the number of defective samples equals, or exceeds, the quantity specified in the "Rejection, Number Defective" column of Table I, the entire lot shall be considered defective.

TABLE I. Sampling plan

Lot size	Sample size	Acceptance Number Defective	Rejection Number Defective
2 to 8	2	0	1
9 to 15	3	0	1
16 to 25	5	0	1
26 to 50	8	1	2
51 to 90	13	1	2
91 to 150	20	2	3
151 to 280	32	3	4
281 to 500	50	5	6
501 to 1200	80	7	8

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Regulatory requirements. The offeror/contractor is encouraged to use recovered materials in accordance with Public Law 94-580 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. The preservation, packaging, packing, labeling, and marking shall be as specified in the contract or order.

Cross reference. The styles and sizes listed in this CID are cross referenced to those listed in specification GGG-S-600 as shown in Table II.

TABLE II. Cross reference

GGG-S-600	A-A-50963
---	Style A - Size 1
---	Style A - Size 2
Pyramid-Size 3/8 inch	Style A - Size 3
Pyramid-Size 5/8 inch	Style A - Size 4
---	Style B - Size 1
Chisel-Size 3/8 inch	Style B - Size 2
Chisel-Size 5/8 inch	Style B - Size 3
Chisel-Size 1 inch	Style B - Size 4

Notes. The purchaser shall specify the quantity of soldering irons and the quantity, style, and size of tips to be furnished in the contract or order. Tips and heating cartridges may be ordered on an individual basis as required for specific procurement.

MILITARY INTERESTS:

Review Activities

Air Force - 99
DLA - GS

CIVIL AGENCY COORDINATING ACTIVITY

GSA - FSS

PREPARING ACTIVITY

DLA - IP

Project 3439-0522

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